

Off-Highway Vehicle Cybersecurity

Secure the Machines Built for the Toughest Environments.

Construction, mining, forestry, material handling, and other off-highway industries are undergoing a digital transformation. Excavators, loaders, cranes, forklifts, autonomous machines, and specialized equipment increasingly depend on electronic control systems, sensors, telematics, connectivity, remote diagnostics, automation, and cloud-based fleet platforms.

As machines become smarter and more connected, their cybersecurity exposure grows.

secureXmotive provides specialized cybersecurity consulting services to help off-highway vehicle manufacturers and technology providers protect machines throughout their complete lifecycle.

Off-highway vehicles operate in environments where availability, safety, productivity, and reliability are critical. A cyberattack that disrupts machine operation, manipulates control functions, compromises telemetry, or disables a fleet management platform can result in significant operational and financial consequences.

Our approach begins by understanding the machine, its operating environment, its architecture, and its connectivity. We identify potential attack paths across ECUs, CAN and Ethernet networks, diagnostic interfaces, wireless connectivity, telematics units, remote-access systems, cloud platforms, and external equipment.

secureXmotive supports manufacturers with **Threat Analysis and Risk Assessment, cybersecurity architecture, security requirements, penetration testing, vulnerability assessment, network security, diagnostic security, secure OTA updates, and cybersecurity lifecycle management.**

We also assess cybersecurity risks associated with increasingly common capabilities such as remote machine monitoring, autonomous operation, predictive maintenance, fleet management, and remote diagnostics.

Our Off-Highway Cybersecurity Services

- Machine and vehicle cybersecurity assessments
- Threat Analysis and Risk Assessment
- Cybersecurity architecture
- ECU and controller security
- CAN, Ethernet and network security
- Telematics and remote-access security
- Diagnostic security
- Penetration testing
- Vulnerability assessment
- OTA update security
- Secure communications
- Fleet cybersecurity

- Autonomous-machine security assessment
- Incident response planning
- Cybersecurity process and governance

From Connected Machines to Secure Fleets

The cybersecurity challenge does not end with the machine. Modern off-highway ecosystems connect machines with dealers, service organizations, cloud platforms, mobile applications, operators, and fleet management systems.

secureXmotive evaluates these interconnected ecosystems to help manufacturers understand where trust boundaries exist, where vulnerabilities could propagate, and how security controls can contain potential attacks.

Our goal is simple: help manufacturers build machines that remain **secure, available, reliable, and resilient** throughout their operational lifetime.

Whether you are developing a new connected machine or modernizing an existing product portfolio, secureXmotive can help embed cybersecurity into your engineering and business processes.

secureXmotive — Securing the Machines That Build, Move, Mine, and Harvest.